MTH645_MID TERM

Lec (1-22)

- LeC1-7 mth645 pg3
- M@l!k MTH645 Important for mid term notes lecture1to15 pg5
- Mth645 solved mid paper (khattak) 14jan pg14
- M@l!k MTH645 Mid term paper pg10
- MTH645 Midterm Paper Fall 21 pg1
- Mth645 mid paper jan2022

MTH645 :- Imp points:-

1- NO of elements of $P(A) = 2^{(\text{no of elements})}$

2- **Properties:-**

- Commutative: $A \cup B = B \cup A$; $A \cap B = B \cap A$
- Associative: $A \cup (B \cup C) = (A \cup B) \cup C$; $(A \cap B) \cap C = A \cap (B \cap C)$
- Idempotency: $A \cup A = A$; $A \cap A = A$
- Identity: $A \cup \phi = A$; $A \cap X = A$; $A \cap \phi = \phi$; $A \cup X = X$
- Transitivity: $A \subset B, B \subset C \Rightarrow A \subset C$
- Involution: $\bar{\bar{A}} = A$; $A \cap \bar{A} = \phi$; $A \cup \bar{A} = X$
- De-morgan's law: $\overline{A \cap B} = \bar{A} \cup \bar{B}$; $\overline{A \cup B} = \bar{A} \cap \bar{B}$

3- **Characteristic Function:-** $(\chi_A(x))$

$$\chi_A(x) = \begin{cases} 1 & \text{if } x \in A \\ 0 & \text{if } x \notin A \end{cases}$$

Set A is written as;

$$A = \{(x, \mu_A(x) \mid x \in A)\}$$

4- **Union and Intersection:-**

$A \cup B = \chi_A(x) \vee \chi_B(x)$; $\max(\chi_A(x), \chi_B(x))$

$A \cap B = \chi_A(x) \wedge \chi_B(x)$; $\min(\chi_A(x), \chi_B(x))$

5- **Complement:-**

$$\bar{A} = 1 - \chi_A(x).$$

6)- When U is finit set then fuzzy set in U rep as=>.

$$A = \sum_{i=1}^{n} \frac{x_i}{\mu_A(x_i)}$$

7)- when U is infinite then=>

$$A = \int_X \frac{x}{\mu_A(x)}$$

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⇒ Expressions of fuzzy set :-

⇒ $A = \{(x, \mu_A(x)\}$

⇒ $A = \frac{\mu_A(x_1)}{x_1} + \frac{\mu_A(x_2)}{x_2} + \cdots + \frac{\mu_A(x_n)}{x_n} + \cdots$

⇒ $\sum_{i=1}^{n} \frac{\mu_A(x_i)}{x_i}$ (For discrete).

⇒ $A = \int \frac{\mu_A(x)}{x}$ (For continuous).

⇒ Degree of subsethood of $\vec{A}$ in $\vec{B}$:-

$$S(\vec{A}, \vec{B}) = \frac{|\vec{A} \cap \vec{B}|}{|\vec{A}|}$$

$S(\vec{A}, \vec{B}) \in [0, 1]$.

⇒ Support of fuzzy set :-

membership values of elements are greater than zero.

⇒ Normal Fuzzy set :-

if ∃ atleast one element whose membership value is one.

⇒ Empty F.S ⇒

the membership value of all elements is zero.

⇒ Cardinality of F.S :- Sum of all membership values of the elements of the set.

⇒ Interval as fuzzy set :-

Expressing the interval as membership function is shown in the following ⇒.

$$\mu_A(x) = \begin{cases} 0 & ; \quad x < a_1 \\ 1 & ; \quad a_1 \le x \le a_3 \\ 0 & ; \quad a_3 < x \end{cases}$$

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⇒ Interval Union:-

$$I_1 \cup I_2 = [a_1, b_1] \cup [a_2, b_2]$$
$$= [\min\{a_1, a_2\}, \max\{b_1, b_2\}]$$

and → $I_1 \cap I_2 \neq \phi$ otherwise it is undefined.

⇒ Interval Intersection:-

$$I_1 \cap I_2 = [a_1, b_1] \cap [a_2, b_2]$$
$$= [\max\{a_1, a_2\}, \min\{b_1, b_2\}]$$

and; $[a_1, b_1] \cap [a_2, b_2] = \phi$ iff $a_1 > b_2$ or $a_2 > b_1$.

⇒ Interval Equality:-

$I_1 = I_2$ if $a_1 = a_2$, $b_1 = b_2$.

⇒ Interval Inequality:-

- $[a_1, b_1] < [a_2, b_2]$ iff $b_1 < a_2$.
- $[a_1, b_1] > [a_2, b_2]$ iff $a_1 > b_2$.

otherwise they can't be compared.

⇒ Interval Inclusion:-

$[a_1, b_1] \subseteq [a_2, b_2]$ iff $a_2 \leq a_1, b_1 \leq b_2$.

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(M@L!k Hafeez 0306-6976635)

Fuzzy set:- A fuzzy set consist of two component. regular set and membership function associated with it. $A = \{(x, \mu_A(x)) / x \in U\}$

Expressions of fuzzy set:-

i) $A = \{(x, \mu_A(x)\}$, ii) $A = \frac{\mu_A(x)}{x_1} + \frac{\mu_A(x_2)}{x_2} + \cdots + \frac{\mu_A(x_n)}{x_n}$

iii) $A = \sum_{i=1}^{n} \frac{\mu_A(x_i)}{x_i}$, iv) $A = \int \frac{\mu_A(x)}{x}$

Theorems:- Vip

i) $(\tilde{A} \cup \tilde{A}') \geq 0.5$ ii) $(\tilde{A} \cap \tilde{A}') \leq 0.5$

Support:- A set is said to be a support of a fuzzy set $\tilde{A}$ if the membership value of the element is greater than zero.

e.g $\tilde{A} = \{(x_1, 0), (x_2, 0.1), (x_3, 0.4)\}$.

support $\tilde{A} = \{x_2, x_3\}$.

Normal fuzzy set:-

A set is said to be a normal fuzzy set if the membership value of the element is equal to one(1).

e.g $\tilde{A} = \{(x_1, 0), (x_2, 0.2), (x_3, 1)\}$.

e^{-x^2} is a normal fuzzy set at $x = 0$.

$e^0 = 1$.

empty fuzzy set:- A set is said to be empty fuzzy set if the membership value of all the elements in the sets are zero.

e.g $\tilde{A} = \{(x_1, 0), (x_2, 0), (x_3, 0)\}$.

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Cardinality:- cardinality of a fuzzy set is the sum of all the membership value of the elements in the set. e.g $\tilde{A} = \{(x_1, 0.9), (x_2, 0.3), (x_3, 1)\}$

$|\tilde{A}| = 0.9 + 0.3 + 1 = 2.2$

Degree of Subsethood:- Degree of subsethood of $\tilde{A}$ in $\tilde{B}$ is defined as $S(\tilde{A}, \tilde{B}) = \frac{|\tilde{A} \cap \tilde{B}|}{|\tilde{A}|}$.

NB

Degree of subsethood always lie in $[0, 1]$.

Union of fuzzy set:- Let $\tilde{F}(x)$ denotes the collection of fuzzy set on the given Universe of discourse X.

Let $\tilde{A}, \tilde{B} \in \tilde{F}(x)$ then union of $\tilde{A}$ and $\tilde{B}$ is the fuzzy set $\tilde{C}$ which is

$\mu_{\tilde{C}}(x) = \max\{\mu_{\tilde{A}}(x), \mu_{\tilde{B}}(x)\}$

$= \mu_{\tilde{A}}(x) \vee \mu_{\tilde{B}}(x) \quad , \forall x \in X.$

Intersection:-

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$\mu_{\tilde{C}}(x) = \min\{\mu_{\tilde{A}}(x), \mu_{\tilde{B}}(x)\}$

$= \mu_{\tilde{A}}(x) \wedge \mu_{\tilde{B}}(x)$

Complement:- Let $\tilde{F}(x)$ denote the collection of fuzzy sets on a given Universe of discourse.

Let $\tilde{A} \in \tilde{F}(x)$ then the complement of $\tilde{A}$ is the fuzzy set $\tilde{B}$ which is

$\mu_{\tilde{B}}(x) = 1 - \mu_{\tilde{A}}(x)$

Also

$\tilde{B} = \overline{\tilde{A}}$

Union of two intervals:- The union of two intervals $I_1 = [a_1, b_1]$ and $I_2 = [a_2, b_2]$ is defined as

$I_1 \cup I_2 = [a_1, b_1] \cup [a_2, b_2]$

$= [\min\{a_1, a_2\}, \max\{b_1, b_2\}]$.

iff $[a_1, b_1] \cap [a_2, b_2] \neq \phi$

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$a_1 \leq b_2$
$b_1 \leq a_2$

Intersection of two intervals:- The intersection of intervals $I_1=(a_1,b_1]$ and $(a_2,b_2]$ is defined as

$$I_1 \cap I_2 = (a_1,b_1] \cap (a_2,b_2]$$
$$= (\max\{a_1,a_2\}, \min\{b_1,b_2\}]$$

and $(a_1,b_1] \cap (a_2,b_2] = \phi$

iff $a_1 > b_2$ ~~and~~ ~~$b_1 \geq a_2$~~ or $a_2 > b_1$.

equality of two intervals:- Two intervals $I_1=(a_1,b_1]$ and $I_2=(a_2,b_2]$ are said to be equal iff $a_1=a_2$ and $b_1=b_2$ denoted by $I_1=I_2$ or $(a_1,b_1]=(a_2,b_2]$

Inequality of Intervals:- Interval $I_1=(a_1,b_1]$ is left then $I_2=(a_2,b_2]$ iff $b_1 < a_2$ denoted by $I_1 < I_2$ or $(a_1,b_1] < (a_2,b_2]$.

- Also two intervals $I_1=(a_1,b_1]$ is said to be greater then $I_2=(a_2,b_2]$ iff $a_1 > b_2$.

(VIP) $\leq$ and $\geq$ are not use in intervals.

Inclusion of intervals:- Interval $I_1=(a_1,b_1]$ is said to be included in $I_2=(a_2,b_2]$ iff $a_1 \geq a_2$ and $b_1 \leq b_2$. More precisely say that I_1 is subset of I_2 denoted by $I_1 \subseteq I_2$.

Triangular Fuzzy Number:- A fuzzy no. which is represent by three points is called triangular fuzzy no.

$$\tilde{A} = (a_1, a_2, a_3)$$

$$\mu_{\tilde{A}}(x) = \begin{cases} 0 & x < a_1 \\ \dfrac{x-a_1}{a_2-a_1} & a_1 \leq x \leq a_2 \\ \dfrac{a_3-x}{a_3-a_2} & \dfrac{a}{2} \leq x \leq a_3 \\ 0 & x > a_3 \end{cases}$$

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Addition of triangular fuzzy No.:- If $\tilde{A}=(a_1,b_1,c_1)$ and $\tilde{B}=(a_2,b_2,c_2)$ be two triangular fuzzy No. then their addition is

$$\tilde{A}+\tilde{B}=(a_1+a_2,\ b_1+b_2,\ c_1+c_2).$$

Negative of triangular fuzzy No.:-

If $\tilde{A}=(a,b,c)$ is a triangular fuzzy No. then $-\tilde{A}=(-c,-b,-a)$ $\quad a\le b\le c$.

Substraction of triangular fuzzy No.:-

If $\tilde{A}=(a_1,b_1,c_1)$ and $\tilde{B}=(a_2,b_2,c_2)$ are triangular fuzzy No. then

$$\begin{aligned}\tilde{A}-\tilde{B}&=(a_1,b_1,c_1)-(a_2,b_2,c_2)\\&=(a_1,b_1,c_1)+(-c_2,-b_2,-a_2)\\&=(a_1-c_2,\ b_1-b_2,\ c_1-a_2)\end{aligned}$$

Multiplication of triangular fuzzy No.:-

$$\begin{aligned}\tilde{A}\times\tilde{B}&=(a_1,b_1,c_1)\times(a_2,b_2,c_2)\\&=(a_1a_2,\ b_1b_2,\ c_1c_2).\end{aligned}$$

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Trapezoidal fuzzy No.:-

A fuzzy No. which is represented by four points is called Trapezoidal fuzzy No.

$$A=(a_1,a_2,a_3,a_4)$$

$$\mu_{\tilde{A}}(x)=\begin{cases}0 & x<a_1\\ \dfrac{x-a_1}{a_2-a_1} & a_1\le x\le a_2\\ 1 & a_2\le x\le a_3\\ \dfrac{a_4-x}{a_4-a_3} & a_3\le x\le a_4\\ 0 & x>a_4\end{cases}$$

Addition of Trapezoidal fuzzy No.:- If $\tilde{A}=(a_1,b_1,c_1,d_1)$ and $\tilde{B}=(a_2,b_2,c_2,d_2)$ are two trapezoidal fuzzy No. then

$$\tilde{A}+\tilde{B}=(a_1+a_2,\ b_1+b_2,\ c_1+c_2,\ d_1+d_2)$$

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Scalar addition of trapezoidal fuzzy No.:-

If k is any scalar then

$k + \tilde{A} = (a_1 + k, b_1 + k, c_1 + k, d_1 + k)$

negative of trapezoidal fuzzy No.:-

If $\tilde{A} = (a_1, b_1, c_1, d_1)$

then $-\tilde{A} = (-d_1, -c_1, -b_1, -a_1)$

Scalar substraction of trapezoidal fuzzy No.:-

If k is any scalar

then $\tilde{A} - k = (a_1 - k, b_1 - k, c_1 - k, d_1 - k)$

Substraction of trapezoidal fuzzy No.:-

$\tilde{A} - \tilde{B} = (a_1, b_1, c_1, d_1) - (a_2, b_2, c_2, d_2)$

$= (a_1, b_1, c_1, d_1) + (-d_2, -c_2, -b_2, -a_2)$

$= (a_1 - d_2, b_1 - c_2, c_1 - b_2, d_1 - a_2)$

Multiplication of Trapezoidal fuzzy No.:-

$\tilde{A} \times \tilde{B} = (a_1 a_2, b_1 b_2, c_1 c_2, d_1 d_2)$

Scalar multiplication of trapezoidal fuzzy No.:-

- if $k \in \mathbb{R}^+$ and k is any scalar

then $\alpha A = (a_1 \alpha, b_1 \alpha, c_1 \alpha, d_1 \alpha)$

- if $k \in \mathbb{R}^-$ then

$\alpha A = (\alpha d_1, \alpha c_1, \alpha b_1, \alpha a_1)$

Inverse of Trapezoidal fuzzy No.:-

If $\tilde{A} = (a, b, c, d)$

then $\tilde{A}^{-1} = \left(\frac{1}{d}, \frac{1}{c}, \frac{1}{b}, \frac{1}{a}\right)$

Division of Trapezoidal Fuzzy No.:-

If $\tilde{A} = (a_1, b_1, c_1, d_1)$ and $\tilde{B} = (a_2, b_2, c_2, d_2)$

then $\tilde{A} / \tilde{B} = \tilde{A} \tilde{B}^{-1}$

$= (a_1, b_1, c_1, d_1) \times \left(\frac{1}{d_2}, \frac{1}{c_2}, \frac{1}{b_2}, \frac{1}{a_2}\right)$

$= \left(\frac{a_1}{d_2}, \frac{b_1}{c_2}, \frac{c_1}{b_2}, \frac{d_1}{a_2}\right)$

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= KHATTAK =

MTH # 645 MID

"SOLVED PAPER"

"KHATTAK"

Q.NO;01 Show $x \Delta_4 x$?.

Sol: As we know that

$$x \Delta_4 y = \frac{xy}{x+y-xy} \quad -(1)$$

By using Eq (1) we write

$$x \Delta_4 x = \frac{xx}{x+x-xx}$$

$$= \frac{x^2}{2x-x^2}$$

$$= \frac{x^2}{x(2-x)}$$

$$x \Delta_4 x = \frac{x}{2-x} \quad \text{Ans.}$$

Q.NO;02 $A = \{10, 15, 20\}$ Find A^2 and show that Graphically ?.

Sol: p.t.o

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on OPPO F5

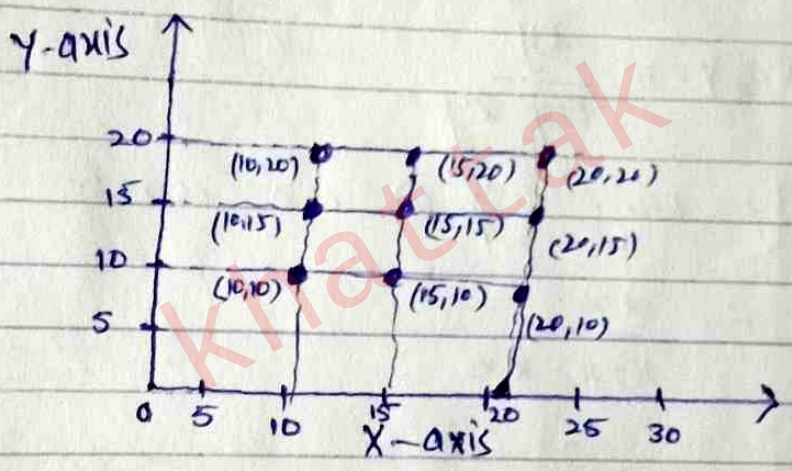
Since Given that $A = \{10, 15, 20\}$

Plan

$A^2 = A \times A = \{10, 15, 20\} \times \{10, 15, 20\}$

$A^2 = \{(10,10), (10,15), (10,20), (15,10), (15,15), (15,20), (20,10), (20,15), (20,20)\}$.

Graphically shown



QNO;03 write the relation for R?

Sol:

	x_1	x_2	x_3
a_1	0	1	0
a_2	1	0	0
a_3	0	1	0
a_4	0	0	1

Sol:

$R = \{(a_1, x_2), (a_2, x_1), (a_3, x_2), (a_4, x_3)\}$

Q NO; 04 Define the intersection of two interval.

Sol: Let Suppose I_1, I_2 be the two interval such that $I_1 = [a_1, b_1]$ and $I_2 = [a_2, b_2]$ than intersection of these two interval can be define As

$$I_1 \cap I_2 = [a_1, b_1] \cap [a_2, b_2]$$

$$= [\max\{a_1, a_2\}, \min\{b_1, b_2\}]$$

and $I_1 \cap I_2 = \phi$ if $a_2 > b_1$ and $a_1 > b_2$

Q NO; 05 Prove that $(\tilde{A} \cap \tilde{A}')(x) \leq 0.5$

Sol:

Proof: As we know that

$$\tilde{A}'(x) = 1 - \tilde{A}(x)$$

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$(\tilde{A}(x)$

now $(\tilde{A} \cap \tilde{A}')(x) = \min(\tilde{A}(x), \tilde{A}'(x))$

$= \min(\tilde{A}(x), 1 - \tilde{A}(x))$.

Here we discuse two cases.

Case 1:-

$\tilde{A}(x) \leq 0.5$

$\Rightarrow 1 - \tilde{A}(x) \geq 0.5$

$\Rightarrow \tilde{A}'(x) \geq 0.5$

But here is intersection we always take minimum

$\Rightarrow (\tilde{A} \cap \tilde{A}')(x) = \min(\tilde{A}(x), \tilde{A}'(x)) \leq 0.5$

$(\tilde{A} \cap \tilde{A}')(x) \leq 0.5$ — (1)

Case 2:- when $\tilde{A}(x) \geq 0.5$

$\Rightarrow 1 - \tilde{A}(x) \leq 0.5$

$\tilde{A}'(x) \leq 0.5$

Same satuation Here we take min.

$\Rightarrow (\tilde{A} \cap \tilde{A}')x = \min(\tilde{A}(x), \tilde{A}'(x)) \leq 0.5$

$(\tilde{A} \cap \tilde{A}')(x) \leq 0.5$ — (2)

Thus from (1) & (2) it clear that $\tilde{A} \cap \tilde{A}' \leq 0.5$ always.

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Q.NO:06 Define empty fuzzy set?

Sol: A fuzzy set is said to be an empty fuzzy set if all the element maps on zero "0" and membership value of all the element is zero.

Example: let $A = \{y_1, y_2, y_3\}$.

$$\mu_A(x) = \{(y_1, 0), (y_2, 0), (y_3, 0)\}.$$

Q.NO:07

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$= x + a$; $x \in X, a \in (Y+Z)$

$=$

Q no 10; 07 Prove that $x + (y+z) = (x+y) + z$

Proof; L.H.S

$x + (y+z) = x + a$; $x \in X, a \in (y+z)$

$= x + y + z$; $x \in X, y \in Y, z \in Z$

$= (x+y) + z$

$= b + z$; $b \in (x+y), z \in Z$

$= (x+y) + z$

$=$ R.H.S

Proved.

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Q NO; 08 FIND $X \Delta_1 X$: ?

Sol. As we know that

$$X \Delta_1 Y = 0 \vee (x+y-1) \quad -(1)$$

By using Eq (1) we write

$$X \Delta_1 X = 0 \vee (x+x-1)$$

$$= 0 \vee (2x-1)$$

$$X \Delta_1 X = 2x-1$$

Q NO; 09 Define Union of two fuzzy set.

Sol:- Let $\tilde{A}$ and $\tilde{B}$ be the two fuzzy set than their union can be define as

$$\mu_{\tilde{A} \cup \tilde{B}} = \mu_{\tilde{C}} = \max \{ \mu_{\tilde{A}}(x), \mu_{\tilde{B}}(x) \}$$

$$= \{ \mu_A(x) \vee \mu_B(x) \} \quad \forall x \in X$$

Where $\tilde{C} = \tilde{A} \cup \tilde{B}$

Q NO; 10 Define degree of subsethood?

Sol:- degree of subsethood of two fuzzy set $\tilde{A}$ and $\tilde{B}$ can be define as

$$S(\tilde{A}, \tilde{B}) = \frac{|\tilde{A} \cap \tilde{B}|}{|\tilde{A}|}$$

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Q.NO;11 Define Normal fuzzy set with example ?

Sol:- A fuzzy set is said to be normal fuzzy set if there exist at least one element whose membership value is one

Example :- $X = \{x_1, x_2, x_3\}$.

$\tilde{A} = \{(x_1, 0.8), (x_2, 0), (x_3, 1)\}$

$\tilde{A}$ is normal fuzzy set.

Q.NO;12 Make the membership table for the fuzzy relation.?

$R = \{((a,c), 0.7), ((b,a), 0.9), ((c,b), 1), ((c,d), 0.2)\}$

Sol

The membership value of the given relation can be written as in matric form.

	a	b	c	d
a	0	0	0.7	0
b	0.9	0	0	0
c	0	1	0	0.2
d	0	0	0	0

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Q NO; 13 Define Support of a fuzzy set with Example. ?

Sol: Support of a fuzzy set define as the set whose all the element have membership value greater than "zero"

Support $\tilde{A} = \{ x : x \in X \text{ and } \tilde{A}(x) > 0 \}$.

Example $X = \{ x_1, x_2, x_3, x_4 \}$

$\tilde{A} = \{ (x_1, 10), (x_2, 0.1), (x_3, 0.5), (x_4, 1) \}$.

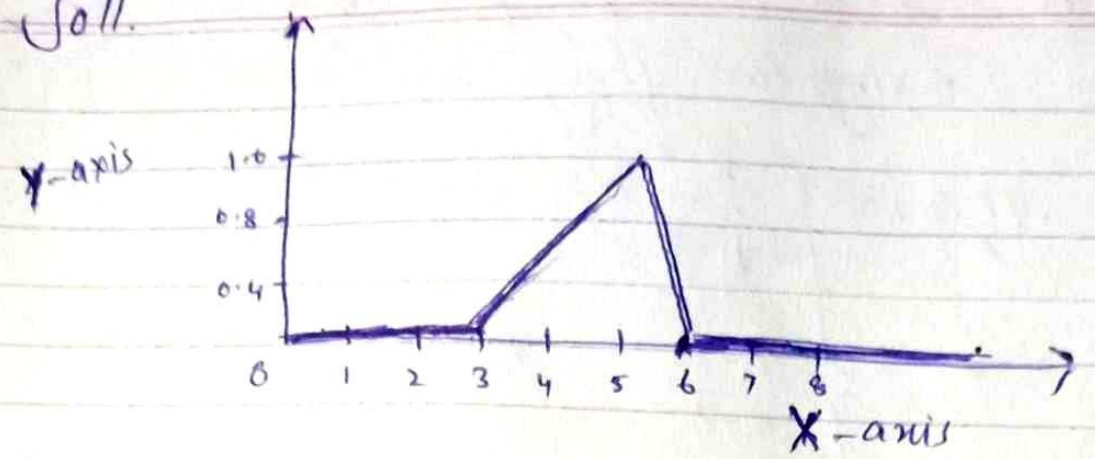
Support $(\tilde{A}) = \{ x_2, x_3, x_4 \}$

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Q NO; 14 Draw the graph of ~~trapezoidal~~ traingular fuzzy no whose membership function as.

$$\mu_A(x) \begin{cases} 0 & ; x < 3 \\ \frac{x-3}{5-3} & ; 3 \le x \le 5 \\ \frac{6-x}{6-5} & ; 5 \le x \le 6 \\ 0 & ; \text{otherwise} \end{cases}$$

Sol.



Q No; 15 If Δ_2 is T-norm and 0.5, 0.4 are two fuzzy number find the value of $0.5 \Delta_2 0.4$

Sol: As we know that Δ_2 T-norm can be defin as.

$$x \Delta_2 y = \frac{xy}{2-(x+y-xy)} \quad \text{—(1)}$$

Here $x = 0.5$, $y = 0.4$

by using Eq (1) we write

$$0.5 \Delta_2 0.4 = \frac{(0.5)(0.4)}{2-(0.5+0.4-(0.5)(0.4))}$$

$$= \frac{0.2}{2-(0.9-0.2)}$$

$$= \frac{0.2}{2-0.7} = \frac{0.2}{1.3}$$

$$0.5 \Delta_2 0.4 = 0.16$$

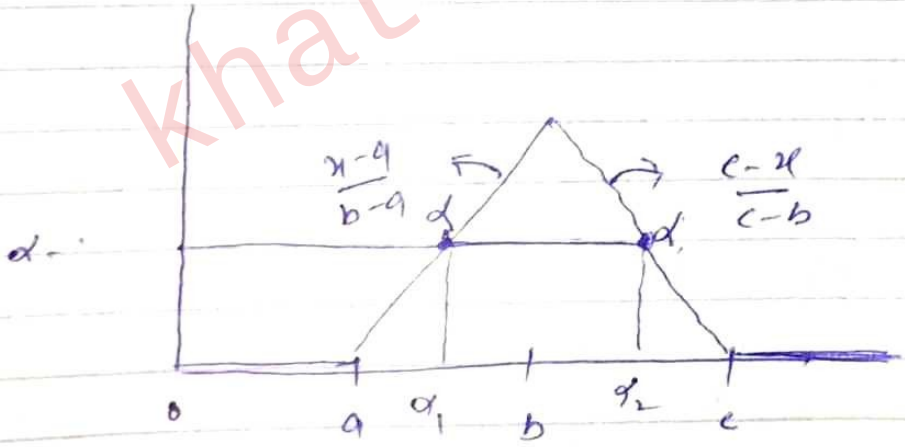
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Q No. 16 membership function of traingular fuzzy no $\tilde{A}$

$$\mu_{\tilde{A}}(x) = \begin{cases} \dfrac{x-a}{b-a} & a \le x \le b \\ \dfrac{c-x}{c-b} & b \le x \le c \\ 0 & \text{otherwise} \end{cases}$$

α-cut of fuzzy number is written as $A_\alpha = (a_1, a_2)$ find a_1, a_2 ?

Sol



This is the graph of traingular fuzzy no $\tilde{A}$.

let $\dfrac{x-a}{b-a} = \alpha$

$\alpha(b-a) = x-a$

$a + \alpha(b-a) = x = a_1$

$\Rightarrow a_1 = a + \alpha(b-a)$

$\dfrac{c-x}{c-b} = \alpha$

$c-x = \alpha(c-b)$

$c - \alpha(c-b) = x = a_2$

$a_2 = c - \alpha(c-b)$

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Q.NO; 17 Define Union of TWO fuzzy set.

Sol: Let $(\tilde{F}, \subseteq)$ be the collection of fuzzy set and $\tilde{A}, \tilde{B}$ belong to $\tilde{F}$ than $\tilde{A} \cup \tilde{B}$ can be define as.

$$\tilde{A} \cup \tilde{B} = \mu_{\tilde{A}}(x) \vee \mu_{\tilde{B}}(x)$$

$$= \max\left(\mu_{\tilde{A}}(x), \mu_{\tilde{B}}(x)\right).$$

Q.NO; 18 Define intersection of two fuzzy set.

Sol: Let $\tilde{F}$ be the collection of fuzzy set in "X" and $\tilde{A}, \tilde{B}$ be the subset of $\tilde{F}$ than their intersection $\tilde{A} \cap \tilde{B}$ can be define As.

$$\tilde{A} \cap \tilde{B} = \mu_{\tilde{A}}(x) \wedge \mu_{\tilde{B}}(x)$$

$$= \min\left(\mu_{\tilde{A}}(x), \mu_{\tilde{B}}(x)\right).$$

khattak

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QNO;19 $\tilde{A} = \{(x_1, 0.1), (x_2, 0.9), (x_3, 1), (x_4, 0), (x_5, 0.3), (x_6, 0.7)\}$

$\tilde{B} = \{(x_1, 0.3), (x_2, 0.8), (x_3, 0), (x_4, 1), (x_5, 0.1), (x_6, 0.9)\}$

Find the degree of subsethood?

Soln

As we know that

$$S(\tilde{A}, \tilde{B}) = \frac{|\tilde{A} \cap B|}{|\tilde{A}|} \quad -(1)$$

$\tilde{A} \cap \tilde{B} = \{(x_1, 0.1), (x_2, 0.8), (x_3, 0), (x_4, 0), (x_5, 0.1), (x_6, 0.7)\}$

$|\tilde{A} \cap \tilde{B}| = 0.1 + 0.8 + 0.1 + 0.7$

$|\tilde{A} \cap \tilde{B}| = 0.17$

$|\tilde{A}| = 0.1 + 0.9 + 1 + 0.3 + 0.7$

$|\tilde{A}| = 1.20$ put in (1)

$$S(\tilde{A}, \tilde{B}) = \frac{0.17}{1.20}$$

$S(\tilde{A}, \tilde{B}) = 0.14$

khatak

Q.

Q.No: 20

M_R	a	b	c
1	0.1	0.7	0
2	0	1	0.9
3	0.5	0	0.2

S_R	a	b	c
1	1	0.8	0.1
2	0.3	0	0.5
3	0.9	1	0

Find MMRUS.

Sol:

MMRUS	a	b	c
1	1	0.8	0.1
2	0.3	1	0.9
3	0.9	1	0.2

khattak

Shot on OPPO F5

0306-6976635
M@L!K Hafeez

MTH - 645

(Mid-Term exam)

Q.No.1 :-

Set:- A set is collection of well defined objects.

e.g $A = \{1, 2, 3, 4, 5\}$ (Set of No.)

$Z = \{0, 1, \ldots\}$ (integers).

Q.No.2)

$x \Delta_4 x = ?$

As we know that if x and y are two fuzzy ~~set~~ No. and Δ_4 is t-norm then

$$x \Delta_4 y = \frac{xy}{x+y-xy}$$

Now

$$x \Delta_4 x = \frac{x \cdot x}{x + x - x \cdot x}$$

$$= \frac{x^2}{2x + x^2}$$

$$= \frac{x^2}{x(2+x)}$$

$$= \frac{x}{2+x} \quad \text{Ans}$$

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Q.No.3:-

Intersection of two intervals:-

Let $I_1 = [a_1, b_1]$ and $I_2 = [a_2, b_2]$ be two intervals then their intersection is given as

$$I_1 \cap I_2 = [\max\{a_1, a_2\}, \min\{b_1, b_2\}]$$

and

$I_1 \cap I_2 = \phi$ iff $a_1 > b_2$ or $a_2 > b_1$

Q.No.4

	x_1	x_2	x_3
q_1	0	1	0
q_2	1	0	0
q_3	0	1	0
q_4	0	0	1

write relation of R.

Sol:-

$R = \{((q_1, x_1), 0), ((q_1, x_2), 1), ((q_1, x_3), 0), ((q_2, x_1), 1), ((q_2, x_2), 0), ((q_2, x_3), 0), ((q_3, x_1), 0), ((q_3, x_2), 1), ((q_3, x_3), 0), ((q_4, x_1), 0), ((q_4, x_2), 0), ((q_4, x_3), 1)\}$

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Q.No.5:-

Prove that $(\tilde{A} \cap \tilde{A}')(x) \leq 0.5$

Proof:-

Consider $\tilde{A}$ be a fuzzy set and $\tilde{A}(x)$ be its membership value of x.

As by definition of complement we have

$$\tilde{A}'(x) = 1 - \tilde{A}(x) \quad \rightarrow (1)$$

So

$$(\tilde{A} \cap \tilde{A}')(x) = \min(\tilde{A}(x), \tilde{A}'(x))$$
$$= \min(\tilde{A}'(x), 1 - \tilde{A}(x))$$

Case I)

If $\tilde{A}(x) \geq 0.5$

then

$$\tilde{A}'(x) = 1 - \tilde{A}(x) \leq 0.5$$

$$\Rightarrow \min(\tilde{A}(x), 1 - \tilde{A}(x)) \leq 0.5$$

Case II

If $\tilde{A}(x) \leq 0.5$

then

$$\tilde{A}'(x) \geq 0.5$$

$$1 - \tilde{A}(x) \geq 0.5$$

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So

$\min(\tilde{A}(x), 1-\tilde{A}(x)) \leq 0.5$

So,

$(\tilde{A} \cap \tilde{A}')(x) \leq 0.5$ (proved)

Q. No. 6:-

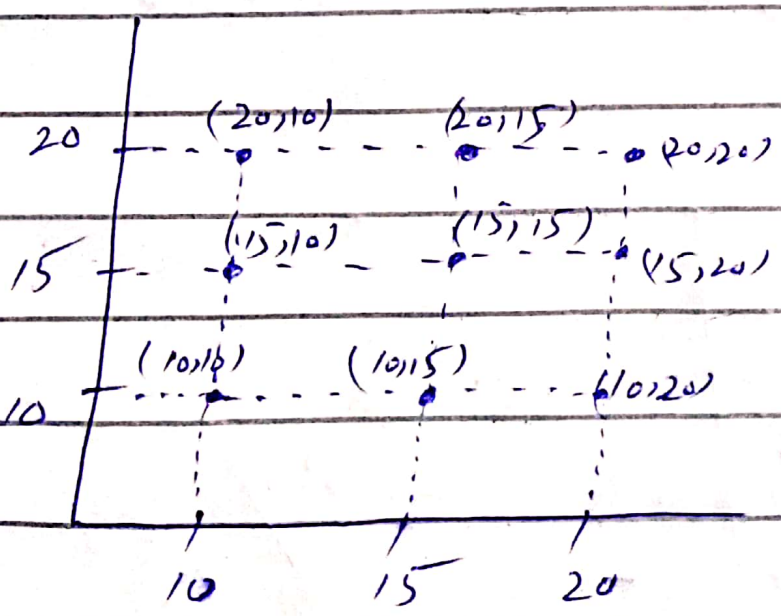
$A = \{10, 15, 20\}$ Find A^2 and show the graphically.

Sol:-

$A = \{10, 15, 20\}$

$A \times A = \{10, 15, 20\} \times \{10, 15, 20\}$

$A^2 = \{(10,10), (10,15), (10,20),$
$(15,10), (15,15), (15,20)$
$(20,10), (20,15), (20,20)\}$.



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Q No. 7):-

Support :- A set is said to be support of a fuzzy set $\tilde{A}$ if the membership value of the element is greater than zero.

i.e

$$\tilde{A} = \{(x_1, 0), (x_2, 0.1), (x_3, 0.4)\}$$

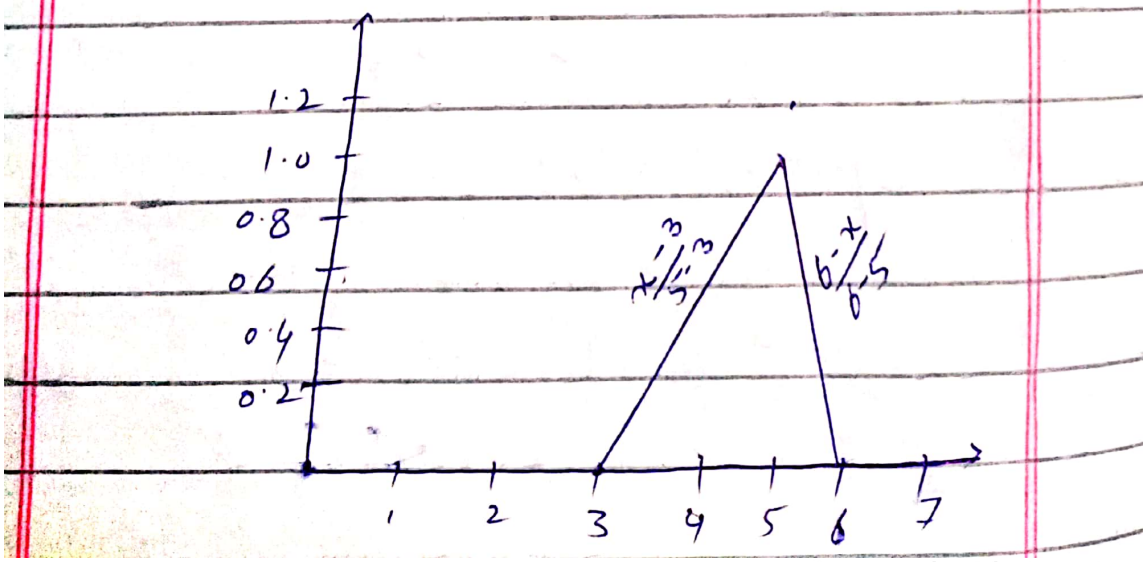
Support of $\tilde{A} = \{x_2, x_3\}$.

Q No. 8:-

Draw graph

$$\mu_A(x) = \begin{cases} \frac{x-3}{5-3} & 3 \le x \le 5 \\ \frac{6-x}{6-5} & 5 \le x \le 6 \\ 0 & \text{otherwise} \end{cases}$$

Sol:-



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Q.No.9

$$\mu_A = \begin{cases} \dfrac{x-a}{b-a} & a \le x \le b \\ \dfrac{c-x}{c-b} & b \le x \le c \\ 0 & \text{otherwise} \end{cases}$$

Find α_1 and α_2 and written as $A_\alpha = [\alpha_1, \alpha_2]$

Sol:-

$$\alpha = \frac{x-a}{b-a}$$

$$\alpha(b-a) = x-a$$

$$\alpha(b-a) + a = x$$

$$\alpha_1 = x = \alpha(b-a)+a$$

and

$$\alpha = \frac{c-x}{c-b}$$

$$\alpha((-b) = c-x$$

$$-x = \alpha((-b) + x$$

$$\alpha_2 = x = c - \alpha((-b)$$

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Now

$$A_\alpha = [\alpha_1, \alpha_2]$$

$$\therefore [\alpha(b-a)+a,\ c-\alpha(c-b)]$$

Q. No. 10):-

Make the membership table for fuzzy relation

$$R = \{((a,c), 0.7), ((b,a), 0.9), ((c,b), 1), ((c,d), 0.2)\}$$

Sol:-

M_R	a	b	c	d
a	0	0	0.7	0
b	0.9	0	0	0
c	0	1	0	0.2

Q. No. 11):-

find $0.8 \Delta_2 0.7$

As we know that if x and y are two fuzzy No. then

$$x \Delta_2 y = \frac{xy}{2-(x+y-xy)}$$

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So

$$0.8 \, \bar{\vee}_2 \, 0.7 = \frac{(0.8) \vee (0.7)}{2 - (0.8 + 0.7 - (0.8) \vee (0.7))}$$

$$= \frac{0.56}{2 - (1.5 - 0.56)}$$

$$= \frac{0.56}{2 - 0.94}$$

Malik

$$= \frac{0.56}{1.06}$$

$$= 0.53$$

(Q. No. 12)

$x \bar{\vee}_1 x$

Ans

$$x \bar{\vee}_1 y = \bar{\vee}(x + y - 1)$$

So

$$x \bar{\vee}_1 x = \bar{\vee}(x + x - 1)$$

$$= \bar{\vee}(2x - 1)$$

(Q. No. 13)

$$X + (Y + Z) = (X + Y) + Z$$

Firstly

$$X + (Y + Z) = \{x + a \mid x \in X, a \in Y + Z\}$$

$$= \{x + (y + z) \mid x \in X, y \in Y, z \in Z\}$$

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$$= \{(x+y)+z \mid x \in X, y \in Y, z \in Z\}$$

$$= \{b+z \mid b \in X+Y, z \in Z\}$$

$$= (X+Y)+Z$$

(Q. No. 14)

M_R	a	b	c
1	0.1	1.0	0.0
2	0.3	1.0	1.0
3	0.0	1.0	1.0

Find Fuzzy Relation.

Sol:-

$$R = \{((1,a),0.1),((1,b),1.0),((1,c),0.0),$$
$$((2,a),0.3),((2,b),1.0),((2,c),1.0),$$
$$((3,a),0.0),((3,b),1.0)((3,c),1.0)\}.$$

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Q.No. 15)

Union of two fuzzy set:-

Let $\tilde{F}(x)$ be the collection of fuzzy sets and $\tilde{A}, \tilde{B} \in \tilde{F}(x)$ then the Union of $\tilde{A}$ and $\tilde{B}$ is the fuzzy set $\tilde{C}$ which is defined as

$$\mu_{\tilde{A} \cup \tilde{B}} = \max \{ \mu_{\tilde{A}}(x), \mu_{\tilde{B}}(x) \}$$

$$= \mu_{\tilde{A}}(x) \vee \mu_{\tilde{B}}(x), \forall x \in X$$

Q.No. 16

Degree of subsethood:-

Degree of subsethood of $\tilde{A}$ in $\tilde{B}$ is defined as

$$S(\tilde{A}, \tilde{B}) = \frac{|\tilde{A} \cap \tilde{B}|}{|\tilde{A}|}$$

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MTH645 – Fuzzy logic and Applications

Midterm Paper Fall 2021

Time: 04:00 PM

Date: 10 – 01 – 2022

Question 01:

Define support group.

Question 02:
In triangular norm Δ_a and 0.5, 0.4 are any fuzzy numbers. Find value of 0.5 Δ_a 0.4.

Question 03:
Let $A = \{10,15,20\}$. Write elements of A^2.

Question 04: Draw graph of triangular fuzzy number whose membership function is

$$\mu_A(x) = \begin{cases} \dfrac{x-3}{5-3} & 3 \le x \le 5 \\ \dfrac{6-x}{6-5} & 5 \le x \le 6 \\ 0 & otherwise \end{cases}$$

Question 05:
Make the membership table for fuzzy relation

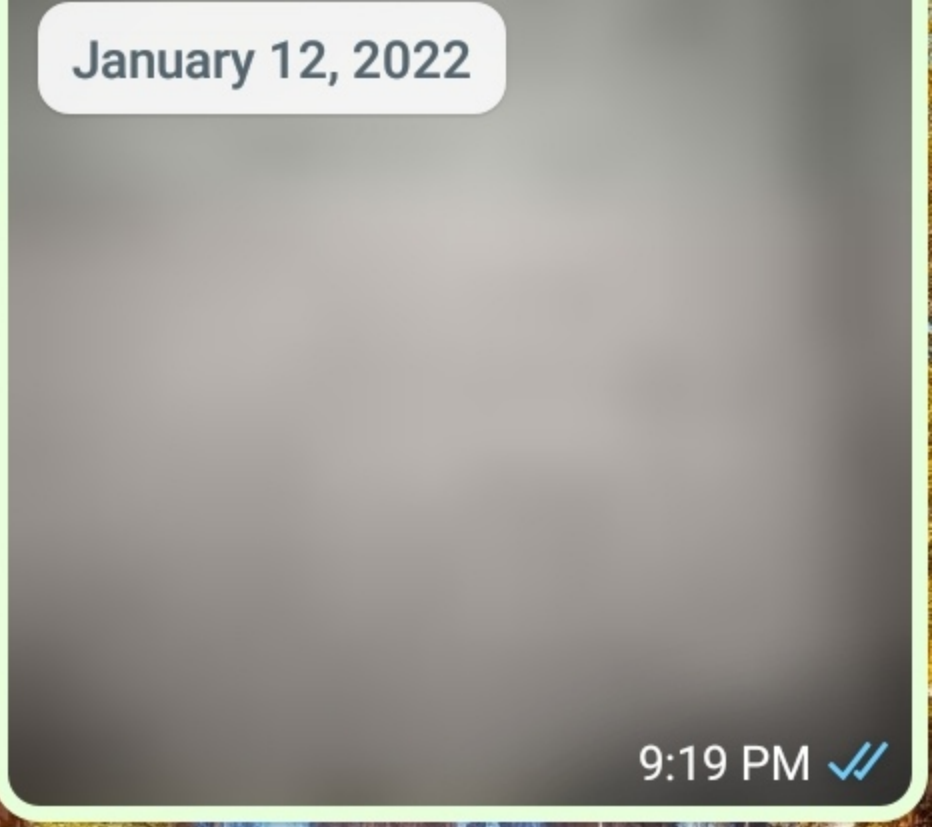
$R = \{((a,c),0.7),((b,a),0.9),((c,b),1),((c,d),0.2)\}$

Question 06:
Membership function of triangular fuzzy number $\tilde{A}$

$$\mu_A = \begin{cases} \dfrac{x-a}{b-a} & a \le x \le b \\ \dfrac{c-x}{c-b} & b \le x \le c \\ 0 & otherwise \end{cases}$$

$\alpha - cut$ Of triangular fuzzy number is written as $A_\alpha = [\alpha_1, \alpha_2]$. *find* α_1 *and* α_2

January 12, 2022



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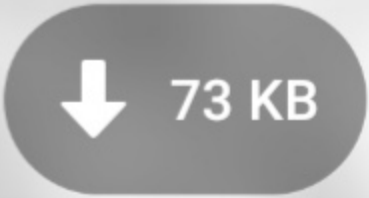
Forwarded

Taday paper mth645
Normal fizzy set with example 3 marks
Susethood degree find krni thi long 5 matks
Values given thi fizzy relation ka table bnana tah. 5 marks
Intersection of two intervals find krna tah 2 marks
T norm se kafi sare mcqs thy baki khan se thy nhi malooooom

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January 14, 2022

January 14, 2022



Mth645

Mcq bhot easy thay aur lectures main say thay aur short question bhi lectures main say thay

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